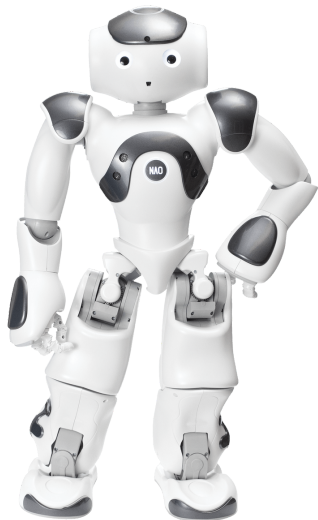


How Autistic Preschool Children Undertake Novel Word Learning from Social Robots

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NAO the social robot

Advertising

Recruitment and consent forms handed out directly to caregivers. If possible, sharing of recruitment forms via newsletters or emails.

Consideration for Children

- Children able to leave at any time
- Social story provided prior to experiment to help with novelty of situation
- Watch a video before participating to minimise any distress
- If a visual schedule is being used, a photograph can be placed on the schedule after discussion with room staff/ center management

Aim

The current research will explore the potential utility of social robots to facilitate word learning with 3–5-year old autistic children. We are interested in how children acquire new words from social robot mediated e-book reading as well as how they engage with the social robot during the various activities.

Procedure

Prior to participating in person, children will be provided with a brief social story about the activity to peruse with their parent or educator.

During the interaction, NAO will run through a pre-programmed script in which the social robot (NAO) introduces itself, invites the child to play with blocks, and asks the child questions before reading the child a short e-book (note: verbal responses are not required to robot questions/prompts and instead explore children's understanding of the story and offers multiple methods for responding such as pointing at pictures).

Children will be presented with a 2-option task to measure their learning and we will explore whether word learning is better than chance (chance being whether children choose answers at random or based off their knowledge obtained from the book read by the social robot.). The study design has previously been done successfully with neurotypical children and has found that the children enjoy the activities.

Requirement of Admin Team and Educators

Staff will be asked to: 1) identify their preferred space for us to conduct the experiment with each child, 2) share ADOS, MSEL and VABS scores for each participating child (where available), 3) store completed consent forms until collected by researcher (~1 week) and 4) collaboratively work out suitable time for testing of each participant around their individual schedules .

Requirement of Caregivers

Caregivers are asked only to read and return the consent form which includes request and consent for the use of existing assessment results (ADOS, MSEL, VABS) of their child. They will be provided with a social story to read to their child prior to participation and a debrief sheet at the end of the experiment.

Requirement of Child

Children's participation in total will take 10–15 minutes and will be conducted at a time to suit centre needs in consultation with the centre manager and room leaders. In the rare chance any individuals dislike or feel discomfort with the robot or situation, they may stop or leave at any time.

Contacts

Ethics approval: [REDACTED] No. [REDACTED]